bradley_spring22: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE's — a formula has none of its own. A line's confidence is not a formula's.

Inline formulas (first occurrence)

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bradley_spring22_F00001	3	1.000	`p`	p	relationship as a simple fraction. If an event occurs with probability p, then we might say the
bradley_spring22_F00002	3	1.000	`1 / p`	$1/p$	amount of information conveyed is $1/p$ because if p is small, then $1/p$ is large and vice versa.
bradley_spring22_F00003	3	1.000	`p=1`	$p=1$	$p = 1$. Since the event is certain to occur, it would not be surprising to learn that it did
bradley_spring22_F00004	3	1.000	`1 / p=1 / 1=1`	$1/p=1/1=1$	$1/p = 1/1 = 1$ is not zero, which goes against that intuition. On the other hand, if $p = 1$, then
bradley_spring22_F00005	3	1.000	`\log (1 / p)=\log (1)=0`	$\log(1/p)=\log(1)=0$	$\log(1/p) = \log(1) = 0$ as desired. This is one reason why logarithms are more convenient.
bradley_spring22_F00006	3	1.000	`\log (1 / p)`	$\log(1/p)$	with probability p to be the number $\log(1/p)$.
bradley_spring22_F00007	3	1.000	`x`	x	nothing special about the numbers 2 and 3 in the previous sentence. If we have *any* two num-
bradley_spring22_F00008	3	1.000	`\log (x)`	$\log(x)$	[1] The logarithm of some number x is another number that we will denote by $\log(x)$ (taken to be the natural logarithm
bradley_spring22_F00009	3	1.000	`\log (1)=0`	$\log(1)=0$	vith n elements. So, entropy is akin to a measure of surprise or u
bradley_spring22_F00010	6	1.000	`2 \times 3=6`	$2\times 3=6$	combine them, say by multiplication, to form a new number: $2 \times 3 = 6$. Of course there is
bradley_spring22_F00011	6	1.000	`y`	y	bers, say x and y, then we can multiply them to obtain a new number $x \times y$, also denoted by
bradley_spring22_F00012	6	1.000	`x \times y`	$x\times y$	bers, say x and y, then we can multiply them to obtain a new number $x \times y$, also denoted by
bradley_spring22_F00013	7	1.000	`x y`	xy	xy. This simple idea of combining things (whether by multiplication or addition or something
bradley_spring22_F00014	7	1.000	`\left(x^{3}\right)^{2} y^{4} x^{-1}`	$\left(x^{3}\right)^{2} y^{4} x^{-1}$	assignments such as "Simplify the expression $(x^3)^2y^4x^{-1}$ using laws of exponents" and "Fac-
bradley_spring22_F00015	7	1.000	`x^{2}+7 x+12^{\prime \prime}`	$x^{2}+7x+12^{\prime\prime}$	tor the quadratic polynomial $x^2 + 7x + 12$" and the like. But this kind of algebra is nothing
bradley_spring22_F00016	7	1.000	`(2 \times 3) \times 5`	$(2\times 3)\times 5$	plied first: $(2 \times 3) \times 5$ is the same as $2 \times (3 \times 5)$, and the analogous holds for concatenation of
bradley_spring22_F00017	7	1.000	`2 \times(3 \times 5)`	$2\times(3\times 5)$	plied first: $(2 \times 3) \times 5$ is the same as $2 \times (3 \times 5)$, and the analogous holds for concatenation of
bradley_spring22_F00018	7	0.985	`A_{\infty}`	A_{∞}	include commutative algebras, associative algebras, Lie algebras, A_∞-algebras, and more.[2] In
bradley_spring22_F00019	8	1.000	`20=19.999 \ldots`	$20=19.999\ldots$	as decimals, and then have to mention aliases like $20 = 19.999\ldots$, a fact not unknown to the
bradley_spring22_F00020	9	0.982	`X^{\prime \prime}`	$X^{\prime\prime}$	Very roughly speaking, "a topology on a set X" is a mathematician's way of declaring which
bradley_spring22_F00021	9	0.983	`X`	X	elements in X are close to each other. When considered together as a pair, both the set and its
bradley_spring22_F00022	10	0.803	`\left(\frac{1}{2}, \frac{1}{2}\right)`	$\left(\frac{1}{2},\frac{1}{2}\right)$	whose sum is 1. For example, $\left(\frac{1}{2},\frac{1}{2}\right)$ and $\left(\frac{2}{5},\frac{1}{2},\frac{1}{10}\right)$ are both probability distributions whereas
bradley_spring22_F00023	10	0.803	`\left(\frac{2}{5}, \frac{1}{2}, \frac{1}{10}\right)`	$\left(\frac{2}{5},\frac{1}{2},\frac{1}{10}\right)$	whose sum is 1. For example, $\left(\frac{1}{2},\frac{1}{2}\right)$ and $\left(\frac{2}{5},\frac{1}{2},\frac{1}{10}\right)$ are both probability distributions whereas
bradley_spring22_F00024	10	1.000	`(7,3)`	$(7,3)$	$(7,3)$ and $\left(-\frac{2}{5},-\frac{1}{2},-\frac{1}{10}\right)$ are not. In the first example, $\frac{1}{2}$ and $\frac{1}{2}$ may represent the respective

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bradley_spring22_ F00025	10	1.000	`\left(-\frac{2}{5},-\frac{1}{2},-\frac{1}{10}\right)`	$\left(-\frac{2}{5},-\frac{1}{2},-\frac{1}{10}\right)$	$(7,3)$ and $\left(-\frac{2}{5},-\frac{1}{2},-\frac{1}{10}\right)$ are not. In the first example, $\frac{1}{2}$ and $\frac{1}{2}$ may represent the respective
bradley_spring22_ F00026	10	1.000	`\frac{1}{2}`	$\frac{1}{2}$	$(7,3)$ and $\left(-\frac{2}{5},-\frac{1}{2},-\frac{1}{10}\right)$ are not. In the first example, $\frac{1}{2}$ and $\frac{1}{2}$ may represent the respective
bradley_spring22_ F00027	10	1.000	`\frac{2}{5}, \frac{1}{2}`	$\frac{2}{5}, \frac{1}{2}$	probabilities of landing a heads or tails on a fair coin toss, while in the second example $\frac{2}{5}, \frac{1}{2}$,
bradley_spring22_ F00028	10	1.000	`\frac{1}{10}`	$\frac{1}{10}$	and $\frac{1}{10}$ may represent the respective probabilities of choosing cereal, oatmeal, or fruit for
bradley_spring22_ F00029	10	0.960	`n`	n	Said more formally, given a **natural number** n (that is, a whole number $1,2,3,\ldots$), a proba-
bradley_spring22_ F00030	10	0.997	`\{1,2, \ldots, n\}`	$\{1,2, \ldots, n\}$	bility distribution on a set $\{1,2,\ldots,n\}$ is a list of nonnegative real numbers $p=(p_1,p_2,\ldots,p_n)$
bradley_spring22_ F00031	10	0.997	`p=\left(p_{1}, p_{2}, \ldots, p_{n}\right)`	$p=\left(p_{1}, p_{2}, \ldots, p_{n}\right)$	bility distribution on a set $\{1,2,\ldots,n\}$ is a list of nonnegative real numbers $p=(p_1,p_2,\ldots,p_n)$
bradley_spring22_ F00032	11	1.000	`p_{1}+p_{2}+\cdots+p_{n}=1`	$p_{1}+p_{2}+\cdots+p_{n}=1$	satisfying $p_1+p_2+\cdots+p_n=1$. Elements in the set $\{1,2,\ldots,n\}$ may be thought of as enu-
bradley_spring22_ F00033	11	0.917	`\left(p_{1}, p_{2}, \ldots, p_{n}\right)`	$\left(p_{1}, p_{2}, \ldots, p_{n}\right)$	The letter p is being used as shorthand to represent the full list of numbers $(p_1,p_2,\ldots,p_n)$.
bradley_spring22_ F00034	11	1.000	`n=2`	$n=2$	Teasing this out with $n=2$, suppose we have a set of two elements $\{1,2\}$ corresponding to
bradley_spring22_ F00035	11	1.000	`\{1,2\}`	$\{1,2\}$	Teasing this out with $n=2$, suppose we have a set of two elements $\{1,2\}$ corresponding to
bradley_spring22_ F00036	11	1.000	`p_{1}=\frac{1}{2}`	$p_{1}=\frac{1}{2}$	the two faces of a coin, namely heads (option #1) or tails (option #2). If we let $p_1=\frac{1}{2}$ and
bradley_spring22_ F00037	11	1.000	`p_{2}=\frac{1}{2}`	$p_{2}=\frac{1}{2}$	$p_2=\frac{1}{2}$, then the list $p=(p_1,p_2)$ is the first example of a probability distribution given above.
bradley_spring22_ F00038	11	1.000	`p=\left(p_{1}, p_{2}\right)`	$p=\left(p_{1}, p_{2}\right)$	$p_2=\frac{1}{2}$, then the list $p=(p_1,p_2)$ is the first example of a probability distribution given above.
bradley_spring22_ F00039	11	1.000	`p=\left(\frac{1}{2}, \frac{1}{2}\right)`	$p=\left(\frac{1}{2}, \frac{1}{2}\right)$	pictures. Figure 2 gives an example. There, the probability distribution $p=\left(\frac{1}{2},\frac{1}{2}\right)$ associated
bradley_spring22_ F00040	11	1.000	`H(p)`	$H(p)$	This number, which we will denote by $H(p)$, is given explicitly in the following formula.
bradley_spring22_ F00041	12	0.966	`\log (1)-\log (p)`	$\log (1)-\log (p)$	of logarithms, this number is the same as $\log(1)-\log(p)$ which is equal to $-\log(p)$ because
bradley_spring22_ F00042	12	0.966	`-\log (p)`	$-\log (p)$	of logarithms, this number is the same as $\log(1)-\log(p)$ which is equal to $-\log(p)$ because
bradley_spring22_ F00043	12	1.000	`-\log \left(p_{1}\right)`	$-\log \left(p_{1}\right)$	first computes the information associated to each event, that is, $-\log(p_1)$ and $-\log(p_2)$ and
bradley_spring22_ F00044	12	1.000	`-\log \left(p_{2}\right)`	$-\log \left(p_{2}\right)$	first computes the information associated to each event, that is, $-\log(p_1)$ and $-\log(p_2)$ and
bradley_spring22_ F00045	12	0.983	`\operatorname{logarithm} \log (p)`	$\operatorname{logarithm} \log (p)$	are important. The natural logarithm $\log(p)$ is negative whenever its input p is between 0 and
bradley_spring22_ F00046	12	1.000	`p=(1,0)`	$p=(1,0)$	to the probability distribution $p=(1,0)$. If we were to learn that this person indeed drank
bradley_spring22_ F00047	12	1.000	`\frac{1}{n}`	$\frac{1}{n}$	comes each with equal probability $\frac{1}{n}$, the entropy of the resulting probability distribution
bradley_spring22_ F00048	12	1.000	`\left(\frac{1}{n}, \frac{1}{n}, \ldots, \frac{1}{n}` `\right)`	$\left(\frac{1}{n}, \frac{1}{n}, \ldots, \frac{1}{n}\right)$	$\left(\frac{1}{n},\frac{1}{n},\ldots,\frac{1}{n}\right)$ will always be $\log(n)$, and this turns out to be the maximum possible value. In
bradley_spring22_ F00049	12	1.000	`\log (n)`	$\log (n)$	$\left(\frac{1}{n},\frac{1}{n},\ldots,\frac{1}{n}\right)$ will always be $\log(n)$, and this turns out to be the maximum possible value. In
bradley_spring22_ F00050	12	1.000	`0 \leq H(p) \leq \log (n)`	$0 \leq H(p) \leq \log (n)$	other words, one can show that $0 \leq H(p) \leq \log(n)$ for any probability distribution p on a set

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bradley_spring22_F00051	13	1.000	`H`	H	We have used the letter H as the name of the function that assigns to a probability distribution
bradley_spring22_F00052	13	1.000	`\Delta_{n}`	Δ_{n}	p its entropy $H(p)$. We will also use Δ_n to denote the set of all possible probability distributions
bradley_spring22_F00053	13	1.000	`\Delta^{n-1}`	Δ^{n-1}	and so on. Mathematicians usually prefer to lower the index by 1 and write Δ^{n-1} instead,
bradley_spring22_F00054	13	1.000	`\mathbb{R}`	$\mathbb{R}$	Δ_n instead.[3] In our menagerie of mathematical notation, we have also used the symbol $\mathbb{R}$ to
bradley_spring22_F00055	13	1.000	`H: \Delta_{n} \rightarrow \mathbb{R}`	$H: \Delta_{n} \rightarrow \mathbb{R}$	the assignment of a probability distribution p to the number $H(p)$. The notation $H\colon \Delta_n \to \mathbb{R}$
bradley_spring22_F00056	13	1.000	`f: X \rightarrow Y`	$f: X \rightarrow Y$	is very convenient, so we will always use analogous notation $f\colon X \to Y$ to mean "a function f
bradley_spring22_F00057	13	1.000	`f`	f	is very convenient, so we will always use analogous notation $f\colon X \to Y$ to mean "a function f
bradley_spring22_F00058	13	1.000	`Y`	Y	from a set X to a set Y that assigns to each input x in X one output $f(x)$ in Y."
bradley_spring22_F00059	13	1.000	`f(x)`	$f(x)$	from a set X to a set Y that assigns to each input x in X one output $f(x)$ in Y."
bradley_spring22_F00060	14	0.979	`n=1,2,3, \ldots`	$n=1,2,3, \ldots$	are infinitely many natural numbers $n = 1,2,3,\ldots$, there are necessarily infinitely many Hs,
bradley_spring22_F00061	14	1.000	`H_{n}`	H_{n}	as well. It would be helpful to indicate this dependency in our notation and write H_n instead
bradley_spring22_F00062	14	1.000	`H: \Delta_{1} \rightarrow \mathbb{R}`	$H: \Delta_{1} \rightarrow \mathbb{R}$	$H\colon \Delta_1 \to \mathbb{R}$ and $H\colon \Delta_2 \to \mathbb{R}$ and $H\colon \Delta_3 \to \mathbb{R}$, and so on.
bradley_spring22_F00063	14	1.000	`H: \Delta_{2} \rightarrow \mathbb{R}`	$H: \Delta_{2} \rightarrow \mathbb{R}$	$H\colon \Delta_1 \to \mathbb{R}$ and $H\colon \Delta_2 \to \mathbb{R}$ and $H\colon \Delta_3 \to \mathbb{R}$, and so on.
bradley_spring22_F00064	14	1.000	`H: \Delta_{3} \rightarrow \mathbb{R}`	$H: \Delta_{3} \rightarrow \mathbb{R}$	$H\colon \Delta_1 \to \mathbb{R}$ and $H\colon \Delta_2 \to \mathbb{R}$ and $H\colon \Delta_3 \to \mathbb{R}$, and so on.
bradley_spring22_F00065	14	1.000	`\Delta_{1}, \Delta_{2}, \Delta_{3}, \ldots`	$\Delta_{1}, \Delta_{2}, \Delta_{3}, \ldots$	portantly, each of these sets $\Delta_1, \Delta_2, \Delta_3, \ldots$ is not merely a set. There is a standard way in
bradley_spring22_F00066	14	1.000	`\Delta_{1}`	Δ_{1}	be shown that Δ_1 is a point, Δ_2 is a line segment, Δ_3 is a triangle, and Δ_4 is a pyramid, as
bradley_spring22_F00067	14	1.000	`\Delta_{2}`	Δ_{2}	be shown that Δ_1 is a point, Δ_2 is a line segment, Δ_3 is a triangle, and Δ_4 is a pyramid, as
bradley_spring22_F00068	14	1.000	`\Delta_{3}`	Δ_{3}	be shown that Δ_1 is a point, Δ_2 is a line segment, Δ_3 is a triangle, and Δ_4 is a pyramid, as
bradley_spring22_F00069	14	1.000	`\Delta_{4}`	Δ_{4}	be shown that Δ_1 is a point, Δ_2 is a line segment, Δ_3 is a triangle, and Δ_4 is a pyramid, as
bradley_spring22_F00070	14	1.000	`n=1`	$n=1$	► *In more detail.* Consider the case when $n = 1$. The set Δ_1 consists of all probability dis-
bradley_spring22_F00071	15	0.985	`x, y`	x, y	numbers (x,y) whose sum is 1; that is, $x+y = 1$. We can rearrange this equation to see it
bradley_spring22_F00072	15	0.985	`x+y=1`	$x+y=1$	numbers (x,y) whose sum is 1; that is, $x+y = 1$. We can rearrange this equation to see it
bradley_spring22_F00073	15	1.000	`y=1-x`	$y=1-x$	is equivalent to $y = 1-x$, which is the equation of a line. Since we have further restricted x
bradley_spring22_F00074	15	1.000	`n=3`	$n=3$	distribution on two elements. Moving on to the case when $n = 3$, the set Δ_3 of probability
bradley_spring22_F00075	15	0.999	`x, y, z`	x, y, z	distributions on three elements is the set of all triples of nonnegative numbers (x,y,z) whose
bradley_spring22_F00076	15	0.922	`x+y+z=1`	$x+y+z=1$	sum is 1; that is $x+y+z = 1$ or equivalently $z = 1-x-y$, which is the equation of a plane
bradley_spring22_F00077	15	0.922	`z=1-x-y`	$z=1-x-y$	sum is 1; that is $x+y+z = 1$ or equivalently $z = 1-x-y$, which is the equation of a plane
bradley_spring22_F00078	15	1.000	`n=4`	$n=4$	case when $n = 4$ requires more work, but it can be shown that the set Δ_4 of all probability

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bradley_spring22_ F00079	15	1.000	`n-1`	$n-1$	the topological space Δ_n is called an $n-1$-**simplex**.[4] For small values of n we have seen that
bradley_spring22_ F00080	17	0.516	`10 \%`	10%	oatmeal, and a 10% chance we choose fruit. This probability distribution on three breakfast
bradley_spring22_ F00081	17	1.000	`q=\left(\frac{2}{5}, \frac{1}{2}, \frac{1}{10}\right)`	$q=\left(\frac{2}{5}, \frac{1}{2}, \frac{1}{10}\right)$	items will be denoted by $q=\left(\frac{2}{5}, \frac{1}{2}, \frac{1}{10}\right)$, which is a point in Δ_3. The picture for this probability
bradley_spring22_ F00082	17	1.000	`r=\left(\frac{3}{10}, \frac{7}{10}\right)`	$r=\left(\frac{3}{10}, \frac{7}{10}\right)$	$r=\left(\frac{3}{10}, \frac{7}{10}\right)$, which is a point in Δ_2. The picture of this probability distribution is shown on
bradley_spring22_ F00083	17	1.000	`q`	q	tribution q on three elements
bradley_spring22_ F00084	17	1.000	`r`	r	larly for r.
bradley_spring22_ F00085	17	0.977	`\frac{1}{2} \times \frac{2}{5}=\frac{1}{5}`	$\frac{1}{2} \times \frac{2}{5}=\frac{1}{5}$	product of the probabilities of each individual outcome, namely $\frac{1}{2} \times \frac{2}{5}=\frac{1}{5}$ or 20%. Similarly,
bradley_spring22_ F00086	17	1.000	`\frac{1}{2} \times \frac{1}{2}=\frac{1}{4}`	$\frac{1}{2} \times \frac{1}{2}=\frac{1}{4}$	equal to $\frac{1}{2} \times \frac{1}{2}=\frac{1}{4}$ or 25%, and the probability of flipping heads and choosing fruit is equal
bradley_spring22_ F00087	17	0.988	`\frac{1}{2} \times \frac{1}{10}=\frac{1}{20}`	$\frac{1}{2} \times \frac{1}{10}=\frac{1}{20}$	to $\frac{1}{2} \times \frac{1}{10}=\frac{1}{20}$ or 5%. This, too, has a visual counterpart. One can imagine grafting the root of
bradley_spring22_ F00088	17	1.000	`\frac{3}{20}`	$\frac{3}{20}$	flipping tails and choosing pizza is $\frac{3}{20}$, and the probability of flipping tails and choosing stir
bradley_spring22_ F00089	17	1.000	`\frac{7}{20}`	$\frac{7}{20}$	fry is $\frac{7}{20}$. The corresponding picture is analogous. To summarize this example, the process of
bradley_spring22_ F00090	17	1.000	`\left(\frac{1}{5}, \frac{1}{4}, \frac{1}{20}, \frac{3}{20}, \frac{7}{20}\right)`	$\left(\frac{1}{5}, \frac{1}{4}, \frac{1}{20}, \frac{3}{20}, \frac{7}{20}\right)$	probability distribution is equal to $\left(\frac{1}{5}, \frac{1}{4}, \frac{1}{20}, \frac{3}{20}, \frac{7}{20}\right)$. It is a point in Δ_5, and its picture is shown
bradley_spring22_ F00091	17	1.000	`\Delta_{5}`	Δ_{5}	probability distribution is equal to $\left(\frac{1}{5}, \frac{1}{4}, \frac{1}{20}, \frac{3}{20}, \frac{7}{20}\right)$. It is a point in Δ_5, and its picture is shown
bradley_spring22_ F00092	18	0.999	`p \times(q, r)`	$p \times(q, r)$	denote this new distribution by, say, "$p \times (q,r)$" to remind us that the probabilities in q and
bradley_spring22_ F00093	20	1.000	`f: \mathbb{R}^{2} \rightarrow \mathbb{R}`	$f: \mathbb{R}^{2} \rightarrow \mathbb{R}$	of the inputs for a different function. Multiplication, for instance, is a function $f\colon \mathbb{R}^2 \to \mathbb{R}$
bradley_spring22_ F00094	20	0.856	`(x, y)`	(x, y)	that accepts a pair of numbers (x,y) as input and computes their product $f(x,y)=xy$ as
bradley_spring22_ F00095	20	0.856	`f(x, y)=x y`	$f(x, y)=x y$	that accepts a pair of numbers (x,y) as input and computes their product $f(x,y)=xy$ as
bradley_spring22_ F00096	20	1.000	`p=`	$p=$	an operation? It is not, really. It would be better to represent a probability distribution $p=$
bradley_spring22_ F00097	20	0.999	`f: \mathbb{R}^{n} \rightarrow \mathbb{R}`	$f: \mathbb{R}^{n} \rightarrow \mathbb{R}$	$(p_1, p_2, \ldots, p_n)$ by an actual (continuous) function $f\colon \mathbb{R}^n \to \mathbb{R}$ that accepts a list of numbers
bradley_spring22_ F00098	20	1.000	`x=\left(x_{1}, x_{2}, \ldots, x_{n}\right)`	$x=\left(x_{1}, x_{2}, \ldots, x_{n}\right)$	$x=(x_1, x_2, \ldots, x_n)$ as input and that outputs a single number $f(x)$. In such a passage from
bradley_spring22_ F00099	20	1.000	`p_{1}, p_{2}, \ldots, p_{n}`	$p_{1}, p_{2}, \ldots, p_{n}$	*probability distributions* to *functions*, the formula for f may involve the probabilities $p_1, p_2, \ldots, p_n$
bradley_spring22_ F00100	20	1.000	`x=`	$x=$	somehow. As an example, given a probability distribution p and a list of arbitrary numbers $x=$
bradley_spring22_ F00101	20	0.995	`\left(x_{1}, x_{2}, \ldots, x_{n}\right)`	$\left(x_{1}, x_{2}, \ldots, x_{n}\right)$	$(x_1, x_2, \ldots, x_n)$ as input, we could define $f(x)$ to be equal to the sum $p_1x_1+p_2x_2+\cdots+p_nx_n$, a
bradley_spring22_ F00102	20	0.995	`p_{1} x_{1}+p_{2} x_{2}+\cdots+p_{n} x_{n}`	$p_{1} x_{1}+p_{2} x_{2}+\cdots+p_{n} x_{n}$	$(x_1, x_2, \ldots, x_n)$ as input, we could define $f(x)$ to be equal to the sum $p_1x_1+p_2x_2+\cdots+p_nx_n$, a
bradley_spring22_ F00103	21	1.000	`p, q`	p, q	p,q and r to obtain $p \circ (q,r)$ in our coin-food example is neither homeless nor isolated in the
bradley_spring22_ F00104	21	1.000	`p \circ(q, r)`	$p \circ(q, r)$	p,q and r to obtain $p \circ (q,r)$ in our coin-food example is neither homeless nor isolated in the
bradley_spring22_ F00105	21	1.000	`H(p \circ(q, r))`	$H(p \circ(q, r))$	tions. In other words, can the formula for $H(p \circ (q,r))$ be reexpressed in terms of $H(p)$ and

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bradley_spring22_F00106	21	1.000	`H(q)`	$H(q)$	$H(q)$ and $H(r)$? Perhaps, for instance, the entropy of the composite distribution is equal to
bradley_spring22_F00107	21	1.000	`H(r)`	$H(r)$	$H(q)$ and $H(r)$? Perhaps, for instance, the entropy of the composite distribution is equal to
bradley_spring22_F00108	21	1.000	`H(p \circ(q, r)) \stackrel{?}{=} H(p)+H(q)+H(r)`	$H(p \circ(q, r)) \stackrel{?}{=} H(p)+H(q)+H(r)$	the sum of the entropies of the individual distributions: $H(p \circ (q,r)) \stackrel{?}{=} H(p)+H(q)+H(r)$.
bradley_spring22_F00109	22	1.000	`q^{1}, q^{2}, \ldots, q^{n}`	$q^{1}, q^{2}, \ldots, q^{n}$	$q^1, q^2, \ldots, q^n$. (Pay careful attention to the difference between subscripts and superscripts:
bradley_spring22_F00110	22	1.000	`p_{1}`	p_{1}	p_1 is a number, whereas q^1 is a list of numbers.) As in our motivating example above, the
bradley_spring22_F00111	22	1.000	`q^{1}`	q^{1}	p_1 is a number, whereas q^1 is a list of numbers.) As in our motivating example above, the
bradley_spring22_F00112	22	1.000	`q^{2}`	q^{2}	q^1 might be an element of Δ_3, while q^2 might be an element of Δ_{17}, while q^3 might be an
bradley_spring22_F00113	22	1.000	`\Delta_{17}`	Δ_{17}	q^1 might be an element of Δ_3, while q^2 might be an element of Δ_{17}, while q^3 might be an
bradley_spring22_F00114	22	1.000	`q^{3}`	q^{3}	q^1 might be an element of Δ_3, while q^2 might be an element of Δ_{17}, while q^3 might be an
bradley_spring22_F00115	22	1.000	`p \circ\left(q^{1}, q^{2}, \ldots, q^{n}\right)`	$p \circ\left(q^{1}, q^{2}, \ldots, q^{n}\right)$	distribution $p \circ (q^1, q^2, \ldots, q^n)$ satisfies the following important equation, which is sometimes
bradley_spring22_F00116	22	1.000	`\left\{F: \Delta_{n} \rightarrow\right.`	$\left\{F: \Delta_{n} \rightarrow\right.$	► *In more detail.* This characterization may be stated more formally as follows: if $\{F\colon \Delta_n \to$
bradley_spring22_F00117	22	1.000	`\mathbb{R}\}`	$\mathbb{R}\}$	$\mathbb{R}\}$ is any other collection of continuous functions satisfying Equation (5), then F must in
bradley_spring22_F00118	22	1.000	`F`	F	$\mathbb{R}\}$ is any other collection of continuous functions satisfying Equation (5), then F must in
bradley_spring22_F00119	22	1.000	`F=c H`	$F=c H$	fact be *equal* to H or to some multiple of it. In symbols this means $F = cH$ for some real
bradley_spring22_F00120	22	1.000	`c`	c	number c. As mentioned above, this important theorem appears in [Lei21] and is actually a
bradley_spring22_F00121	23	1.000	`\left\{F: \Delta_{n} \rightarrow \mathbb{R}\right\}_{n \geq 1}`	$\left\{F: \Delta_{n} \rightarrow \mathbb{R}\right\}_{n \geq 1}$	**Theorem 1.** *Let $\{F\colon \Delta_n \to \mathbb{R}\}_{n\geq 1}$ be a sequence of functions. The following are equivalent:*
bradley_spring22_F00122	23	1.000	`H(p \circ q)=H(p)+H(q)`	$H(p \circ q)=H(p)+H(q)$	form $H(p \circ q) = H(p) + H(q)$ for probability distributions p and q. In words, this equation
bradley_spring22_F00123	23	0.998	`\circ`	$\circ$	seems[5] to say that entropy H behaves nicely with composition $\circ$ of probability distributions
bradley_spring22_F00124	23	0.641	`f(\bullet \bullet)=f(\bullet) f(\bullet)`	$f(\bullet \bullet)=f(\bullet) f(\bullet)$	Y; that is, if the function satisfies the equation $f(\bullet\bullet) = f(\bullet)f(\bullet)$ for all elements $\bullet$ and $\bullet$ in X.
bradley_spring22_F00125	23	0.641	`\bullet`	$\bullet$	Y; that is, if the function satisfies the equation $f(\bullet\bullet) = f(\bullet)f(\bullet)$ for all elements $\bullet$ and $\bullet$ in X.
bradley_spring22_F00126	23	1.000	`p \circ q=q`	$p \circ q=q$	$p \circ q = q$ and $H(p) = 0$ and so the chain rule reduces to $H(q) = H(q)$, which is uninteresting. But we will momentarily
bradley_spring22_F00127	23	1.000	`H(p)=0`	$H(p)=0$	$p \circ q = q$ and $H(p) = 0$ and so the chain rule reduces to $H(q) = H(q)$, which is uninteresting. But we will momentarily
bradley_spring22_F00128	23	1.000	`H(q)=H(q)`	$H(q)=H(q)$	$p \circ q = q$ and $H(p) = 0$ and so the chain rule reduces to $H(q) = H(q)$, which is uninteresting. But we will momentarily
bradley_spring22_F00129	24	1.000	`q^{1}=q`	$q^{1}=q$	[6] Notice this equation coincides with Equation (4) when $p = \left(\frac{1}{2}, \frac{1}{2}\right)$ corresponds to a coin toss and when $q^1 = q$ and
bradley_spring22_F00130	24	1.000	`q^{2}=r`	$q^{2}=r$	$q^2 = r$ correspond to our breakfast and dinner choices in the example in the previous section.
bradley_spring22_F00131	25	1.000	`f: \mathbb{R} \rightarrow \mathbb{R}`	$f: \mathbb{R} \rightarrow \mathbb{R}$	functions. To elaborate, the derivative of a differentiable function $f\colon \mathbb{R} \to \mathbb{R}$ (that is, a function
bradley_spring22_F00132	25	1.000	`f^{\prime}`	$f^{\prime}$	whose derivative exists) is another function often denoted by f' or by $\frac{df}{dx}$ or by $d(f)$. Given
bradley_spring22_F00133	25	1.000	`\frac{d f}{d x}`	$\frac{d f}{d x}$	whose derivative exists) is another function often denoted by f' or by $\frac{df}{dx}$ or by $d(f)$. Given

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bradley_spring22_F00134	25	■1.000	`d(f)`	$d(f)$	whose derivative exists) is another function often denoted by f' or by $\frac{df}{dx}$ or by $d(f)$. Given
bradley_spring22_F00135	25	■1.000	`g`	g	two differentiable functions f and g, it is natural to ask about the derivative of their product.
bradley_spring22_F00136	25	■1.000	`(f g)^{\prime}`	$(f g)^{\prime}$	Can the derivative $(fg)'$ be expressed in terms of the individual functions f' and g'? The
bradley_spring22_F00137	25	■1.000	`g^{\prime}`	$g^{\prime}$	Can the derivative $(fg)'$ be expressed in terms of the individual functions f' and g'? The
bradley_spring22_F00138	25	■0.999	`(f g)^{\prime}(x)=f^{\prime}(x) g(x)+f(x) g^{\prime}(x)`	$(f g)^{\prime}(x)=f^{\prime}(x) g(x)+f(x) g^{\prime}(x)$	the function whose value at a point x is given by $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$. In words,
bradley_spring22_F00139	25	■0.978	`f g`	$f g$	the derivative of fg is obtained by multiplying g by the derivative of f, then multiplying f by
bradley_spring22_F00140	25	■1.000	`d(f g)=d(f) g+f d(g)`	$d(f g)=d(f) g+f d(g)$	succinctly as $d(fg) = d(f)g + fd(g)$, which should be compared with Equation (8).
bradley_spring22_F00141	25	■1.000	`d`	d	once again intentional. Then, informally speaking, a **derivation** is defined to be any function d
bradley_spring22_F00142	25	■0.322	`d(\bullet \bullet)=d(\bullet) \bullet+\bullet d(\bullet)`	$d(\bullet \bullet)=d(\bullet) \bullet+\bullet d(\bullet)$	on this set that satisfies the Leibniz rule $d(\bullet\bullet) = d(\bullet)\bullet + \bullet d(\bullet)$ for all elements $\bullet$ and $\bullet$. As one
bradley_spring22_F00143	25	■1.000	`A`	A	makes sense. Given an algebra A, for instance, a derivation on A is formally defined to be a
bradley_spring22_F00144	25	■0.737	`[0,1]`	$[0,1]$	[7] As an example, here is a simple exercise. Let $[0,1]$ denote the set of all numbers between and including 0 and 1, and
bradley_spring22_F00145	25	■1.000	`d:[0,1] \rightarrow \mathbb{R}`	$d:[0,1] \rightarrow \mathbb{R}$	define a function $d\colon [0,1] \to \mathbb{R}$ by declaring $d(x) = -x\log(x)$ if $x > 0$ and $d(x) = 0$ if $x = 0$. Show that d satisfies
bradley_spring22_F00146	25	■1.000	`d(x)=-x \log (x)`	$d(x)=-x \log (x)$	define a function $d\colon [0,1] \to \mathbb{R}$ by declaring $d(x) = -x\log(x)$ if $x > 0$ and $d(x) = 0$ if $x = 0$. Show that d satisfies
bradley_spring22_F00147	25	■1.000	`x>0`	$x>0$	define a function $d\colon [0,1] \to \mathbb{R}$ by declaring $d(x) = -x\log(x)$ if $x > 0$ and $d(x) = 0$ if $x = 0$. Show that d satisfies
bradley_spring22_F00148	25	■1.000	`d(x)=0`	$d(x)=0$	define a function $d\colon [0,1] \to \mathbb{R}$ by declaring $d(x) = -x\log(x)$ if $x > 0$ and $d(x) = 0$ if $x = 0$. Show that d satisfies
bradley_spring22_F00149	25	■1.000	`x=0`	$x=0$	define a function $d\colon [0,1] \to \mathbb{R}$ by declaring $d(x) = -x\log(x)$ if $x > 0$ and $d(x) = 0$ if $x = 0$. Show that d satisfies
bradley_spring22_F00150	25	■0.958	`d(x y)=d(x) y+x d(y)`	$d(x y)=d(x) y+x d(y)$	the Leibniz rule. That is, show that $d(xy) = d(x)y + xd(y)$ for all numbers x and y in $[0,1]$. This computation only
bradley_spring22_F00151	26	■1.000	`d: A \rightarrow A`	$d: A \rightarrow A$	linear function $d\colon A \to A$ satisfying the Leibniz rule $d(ab) = d(a)b + ad(b)$ for all elements
bradley_spring22_F00152	26	■1.000	`d(a b)=d(a) b+a d(b)`	$d(a b)=d(a) b+a d(b)$	linear function $d\colon A \to A$ satisfying the Leibniz rule $d(ab) = d(a)b + ad(b)$ for all elements
bradley_spring22_F00153	26	■1.000	`a`	a	a and b in A. This can be generalized slightly by replacing the target A with another object
bradley_spring22_F00154	26	■1.000	`b`	b	a and b in A. This can be generalized slightly by replacing the target A with another object
bradley_spring22_F00155	26	■0.996	`M`	M	M called a "bimodule over A" and instead considering functions $d\colon A \to M$ satisfying the
bradley_spring22_F00156	26	■0.996	`A^{\prime \prime}`	$A^{\prime \prime}$	M called a "bimodule over A" and instead considering functions $d\colon A \to M$ satisfying the
bradley_spring22_F00157	26	■0.996	`d: A \rightarrow M`	$d: A \rightarrow M$	M called a "bimodule over A" and instead considering functions $d\colon A \to M$ satisfying the
bradley_spring22_F00158	26	■1.000	`d(a)`	$d(a)$	same equation. These, too, are called derivations. The only difference now is that $d(a)$ is an
bradley_spring22_F00159	26	■1.000	`d(a) b`	$d(a) b$	of the expressions $d(a)b$ and $ad(b)$. Here, $d(a)$ and b are elements of different sets, namely
bradley_spring22_F00160	26	■1.000	`a d(b)`	$a d(b)$	of the expressions $d(a)b$ and $ad(b)$. Here, $d(a)$ and b are elements of different sets, namely
bradley_spring22_F00161	26	■1.000	`0.5 \times 8=4`	$0.5 \times 8=4$	members of the same set. By way of analogy, multiplication of two numbers $0.5 \times 8 = 4$ makes
bradley_spring22_F00162	26	■0.960	`0.5 \times`	$0.5 \times$	sense, but what would it mean to multiply a number with an English word, $0.5 \times \textit{yellow} = ?$

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bradley_spring22_ F00163	26	1.000	`d(b)`	$d(b)$	and $d(b)$. Bimodules are the answer to such questions. A **bimodule over** A is a mathematical
bradley_spring22_ F00164	26	0.848	`\mathbf{v}`	$\mathbf{v}$	multiplication. The ability to multiply a vector **v** by a real number k to obtain a new vector
bradley_spring22_ F00165	26	0.848	`k`	k	multiplication. The ability to multiply a vector **v** by a real number k to obtain a new vector
bradley_spring22_ F00166	27	1.000	`\left\{H: \Delta_{n} \rightarrow \mathbb{R}\right\}`	$\left\{H: \Delta_{n} \rightarrow \mathbb{R}\right\}$	of infinitely many continuous functions $\{H\colon \Delta_n \to \mathbb{R}\}$ satisfying a certain equation—the chain
bradley_spring22_ F00167	27	0.947	`\left\{d: \Delta_{n} \rightarrow \boldsymbol{-}\right\}`	$\left\{d: \Delta_{n} \rightarrow \boldsymbol{-}\right\}$	finitely many continuous functions $\{d\colon \Delta_n \to \blacksquare\}$ satisfying a certain equation—the Leibniz
bradley_spring22_ F00168	27	1.000	`d(p): \mathbb{R}^{n} \rightarrow \mathbb{R}`	$d(p): \mathbb{R}^{n} \rightarrow \mathbb{R}$	operad of topological simplices assigns a *continuous function* $d(p)\colon \mathbb{R}^n \to \mathbb{R}$ to each probability
bradley_spring22_ F00169	27	1.000	`\mathbb{R}^{n}`	$\mathbb{R}^{n}$	black box above. It is precisely the set of all continuous functions from $\mathbb{R}^n$ to $\mathbb{R}$. This set can
bradley_spring22_ F00170	27	0.921	`\operatorname{hom}\left(\mathbb{R}^{n}, \mathbb{R}\right)`	$\operatorname{hom}\left(\mathbb{R}^{n}, \mathbb{R}\right)$	Even so, let $\hom(\mathbb{R}^n, \mathbb{R})$ denote this topological space of functions. Then we can summarize
bradley_spring22_ F00171	27	1.000	`\left\{d: \Delta_{n} \rightarrow \operatorname{hom}\left(\mathbb{R}^{n}, \mathbb{R}\right)\right\}`	$\left\{d: \Delta_{n} \rightarrow \operatorname{hom}\left(\mathbb{R}^{n}, \mathbb{R}\right)\right\}$	*continuous functions* $\{d\colon \Delta_n \to \hom(\mathbb{R}^n, \mathbb{R})\}$ *that satisfies an appropriate version of the Leibniz rule,*
bradley_spring22_ F00172	27	1.000	`d(p \circ q)=d(p) \circ q+q \circ d(q)`	$d(p \circ q)=d(p) \circ q+q \circ d(q)$	*"$d(p \circ q) = d(p) \circ q + q \circ d(q)$" for any probability distributions p and q.*
bradley_spring22_ F00173	27	1.000	`p \circ q`	$p \circ q$	The expression $p \circ q$ does not make much sense, for instance. So far we have only used the
bradley_spring22_ F00174	27	1.000	`d(p) \circ q`	$d(p) \circ q$	also need to make sense of the expressions $d(p) \circ q$ and $q \circ d(q)$ that appear in the formula.
bradley_spring22_ F00175	27	1.000	`q \circ d(q)`	$q \circ d(q)$	also need to make sense of the expressions $d(p) \circ q$ and $q \circ d(q)$ that appear in the formula.
bradley_spring22_ F00176	27	1.000	`d(q)`	$d(q)$	probability distribution, whereas $d(q)$ is a function. What does it mean to combine the two?
bradley_spring22_ F00177	27	1.000	`\mathbb{R}^{n} \rightarrow \mathbb{R}`	$\mathbb{R}^{n} \rightarrow \mathbb{R}$	[8] Formally speaking, we can equip the set of continuous functions $\mathbb{R}^n \to \mathbb{R}$ with what is known as the "product
bradley_spring22_ F00178	28	1.000	`d\left(p \circ\left(q^{1}, q^{2}, \ldots, q^{n}\right)\right)`	$d\left(p \circ\left(q^{1}, q^{2}, \ldots, q^{n}\right)\right)$	the chain rule essentially says that the function $d(p \circ (q^1, q^2, \ldots, q^n))$ is obtained by applying
bradley_spring22_ F00179	29	1.000	`d(p)(x)=H(p)`	$d(p)(x)=H(p)$	the function $d(p)\colon \mathbb{R}^n \to \mathbb{R}$ to be constant at entropy; that is, define $d(p)(x) = H(p)$ for all
bradley_spring22_ F00180	29	1.000	`d(p)(x)=c H(p)`	$d(p)(x)=c H(p)$	exists a point x in $\mathbb{R}^n$ so that $d(p)(x) = cH(p)$ for some real number c. The proof of this part
bradley_spring22_ F00181	29	1.000	`d(p)(0)=c H(p)`	$d(p)(0)=c H(p)$	easily show that the special point x is actually zero, that is, $d(p)(0) = cH(p)$.